

1. Obtener las bases de datos creadas. “**show dbs**”

```
>_ mongosh: local +
>_MONGOSH
> show dbs
< PSOE          72.00 KiB
  admin         40.00 KiB
  cine_db       80.00 KiB
  config        108.00 KiB
  local         72.00 KiB
  sample_airbnb 52.52 MiB
  sample_analytics 9.55 MiB
  sample_geospatial 1008.00 KiB
  sample_guides 56.00 KiB
  sample_mflix  94.73 MiB
  sample_restaurants 5.93 MiB
  sample_supplies 992.00 KiB
  sample_training 41.00 MiB
  sample_weatherdata 2.41 MiB
test>
```

2. Sobre la base de datos `sample_training`
 1. Recuperar todas las colecciones “**show collections**”

```
>_MONGOSH ⓘ
> show collections
< companies
  grades
  inspections
  posts
  routes
  trips
  zips
sample_training>
```

2. Recuperar qué base de datos está activa. “**db**”

```
>_ mongosh: local +
>_MONGOSH
> db
< sample_training
sample_training>
```

3. Sobre la colección `sample_training.zip`:

1. Recuperar todos los documentos. “**db.zip.find()**”

```
>_ mongosh: local +
>_MONGOSH
> db
< sample_training
> db.zip.find()
< {
  _id: ObjectId('5c8eccc1caa187d17ca6ed16'),
  city: 'ALPINE',
  zip: '35014',
  loc: {
    y: 33.331165,
    x: 86.208934
  },
  pop: 3062,
  state: 'AL'
}
```

2. Recuperar el primer documento.

“db.zips.findOne()”

```
> db.zips.findOne()
< {
  _id: ObjectId('5c8eccc1caa187d17ca6ed16'),
  city: 'ALPINE',
  zip: '35014',
  loc: {
    y: 33.331165,
    x: 86.208934
  },
  pop: 3062,
  state: 'AL'
}
sample_training> |
```

3. Recuperar los documentos de la ciudad de **SAN DIEGO**.

“db.zips.find({city: “SAN DIEGO”})”

```
> db.zips.find({city: "SAN DIEGO"})
< {
  _id: ObjectId('5c8eccc1caa187d17ca6f4b5'),
  city: 'SAN DIEGO',
  zip: '92101',
  loc: {
    y: 32.71852,
    x: 117.159316
  },
  pop: 20265,
  state: 'CA'
}
```

4. Recuperar los códigos postales (`zip`) cuya ciudad sea `SAN DIEGO` y el estado sea `CA`.

“`db.zips.find({city: "SAN DIEGO", state: "CA"}, {zip: 1})`”

```
> db.zips.find({city: "SAN DIEGO", state: "CA"}, {zip: 1})
< {
  _id: ObjectId('5c8eccc1caa187d17ca6f4b5'),
  zip: '92101'
}
{
  _id: ObjectId('5c8eccc1caa187d17ca6f4b7'),
  zip: '92102'
}
{
  _id: ObjectId('5c8eccc1caa187d17ca6f4b9'),
  zip: '92103'
}
{
  _id: ObjectId('5c8eccc1caa187d17ca6f4bb'),
  zip: '92104'
}
{
  _id: ObjectId('5c8eccc1caa187d17ca6f4bc'),
  zip: '92105'
}
```

5. Recuperar los documentos cuya ciudad sea `SAN DIEGO` pero no pertenezcan al estado de `CA`.

“`db.zips.find({city: "SAN DIEGO", state: { $ne: "CA"}})`”

```
> db.zips.find({city: "SAN DIEGO", state: { $ne: "CA"}})
< {
  _id: ObjectId('5c8eccc1caa187d17ca75196'),
  city: 'SAN DIEGO',
  zip: '78384',
  loc: {
    y: 27.76535,
    x: 98.250297
  },
  pop: 5018,
  state: 'TX'
}
sample_training>
```

6. Recuperar los documentos cuya ciudad sea **SAN DIEGO** y el estado sea **CA** o **TX**.

“db.zips.find({city: "SAN DIEGO", state: { \$in: ["CA", "TX"]}})”

```
> db.zips.find({city: "SAN DIEGO", state: { $in: ["CA", "TX"]}})
< {
  _id: ObjectId('5c8eccc1caal87d17ca6f4b5'),
  city: 'SAN DIEGO',
  zip: '92101',
  loc: {
    y: 32.71852,
    x: 117.159316
  },
  pop: 20265,
  state: 'CA'
}
{
  _id: ObjectId('5c8eccc1caal87d17ca6f4b7'),
  city: 'SAN DIEGO',
  zip: '92102',
  loc: {
    y: 32.713893,
    x: 117.121858
  },
  pop: 45265,
  state: 'CA'
}
```

7. Recuperar los documentos cuya ciudad sea **SAN DIEGO** o el estado sea **TX**.

**“db.zips.find({ \$or:
[{ city: "SAN DIEGO" }, { state: "TX" }] })”**

```
>_MONGOSH  
  
> db.zips.find({  
  $or: [{ city: "SAN DIEGO" }, { state: "TX" }] })  
< {  
  _id: ObjectId('5c8eccc1caa187d17ca6f4b5'),  
  city: 'SAN DIEGO',  
  zip: '92101',  
  loc: {  
    y: 32.71852,  
    x: 117.159316  
  },  
  pop: 20265,  
  state: 'CA'  
}  
{  
  _id: ObjectId('5c8eccc1caa187d17ca6f4b7'),  
  city: 'SAN DIEGO',  
  zip: '92102',  
  loc: {  
    y: 32.713893,  
    x: 117.121858  
  },  
  pop: 45265,  
  state: 'CA'  
}
```

8. Recuperar los documentos donde el `zip` sea 92135, 92136 o 92137.
“`db.zips.find({ zip: { $in: ["92135", "92136", "92137"] } })`” “

```
>_MONGOSH
> db.zips.find({ zip: { $in: ["92135", "92136", "92137"] } })
< {
  _id: ObjectId('5c8eccc1caa187d17ca6f4d7'),
  city: 'SAN DIEGO',
  zip: '92136',
  loc: {
    y: 32.681585,
    x: 117.124678
  },
  pop: 11750,
  state: 'CA'
}
{
  _id: ObjectId('5c8eccc1caa187d17ca6f4e9'),
  city: 'SAN DIEGO',
  zip: '92135',
  loc: {
    y: 32.702482,
    x: 117.19202
  },
  pop: 8122,
  state: 'CA'
}
sample_training>|
```

9. ¿Cuál es el código postal de la ciudad de ALLEN que no tiene habitantes (sólo recupera el `zip`, no nos interesa ningún otro campo, ni el `_id`)?
“`db.zips.find({city: "ALLEN", pop: 0}, {zip: 1, _id: 0})`”

```
> db.zips.find({city: "ALLEN", pop: 0}, {zip: 1, _id: 0})
< {
  zip: '36419'
}
sample_training>
```

